Specification Driven Development (SDD) Agent – Developer Guide

This guide explains how to use the **Specification Driven Development (SDD) Agent** within the MiDAS GitHub workflow to orchestrate complete, end-to-end feature development.

Introduction

The Specification Driven Development (SDD) Agent is a master orchestration agent designed to manage the complete software development lifecycle. It guides a feature from requirements analysis through design, planning, implementation, testing, code review, documentation, and deployment readiness.

Unlike single-purpose agents, the SDD Agent dynamically determines which phases are required based on the user's request. It coordinates specialized sub-agents to ensure high-quality deliverables.

The goal of the SDD Agent is to enforce structured, specification-first development across the MiDAS platform.

What Are the Use Cases for the SDD Agent

The SDD Agent should be used when a feature requires structured lifecycle execution rather than isolated actions.

Common Use Cases

- Building a complete feature from scratch
- Executing structured development from specification to deployment
- Orchestrating multiple phases (design + implementation + testing)

- Validating quality before release
- Managing cross-subsystem changes
- Driving large or complex enhancements

Response Mode Selection

The SDD Agent intelligently selects workflow scope based on the user request.

Full Workflow

- Used when a complete feature request or specification is provided.
- Executes all lifecycle phases sequentially.

Partial Workflow

- Used when some phases are already completed.
- Skips completed stages and continues from the appropriate phase.

Single Phase Execution

- Used when explicitly requested.
- Example:
- “Just design the API” → Only Phase 2
- “Implement from this design” → Start at Phase 4
- “Review this code” → Jump to Phase 6

The agent clarifies scope before execution.

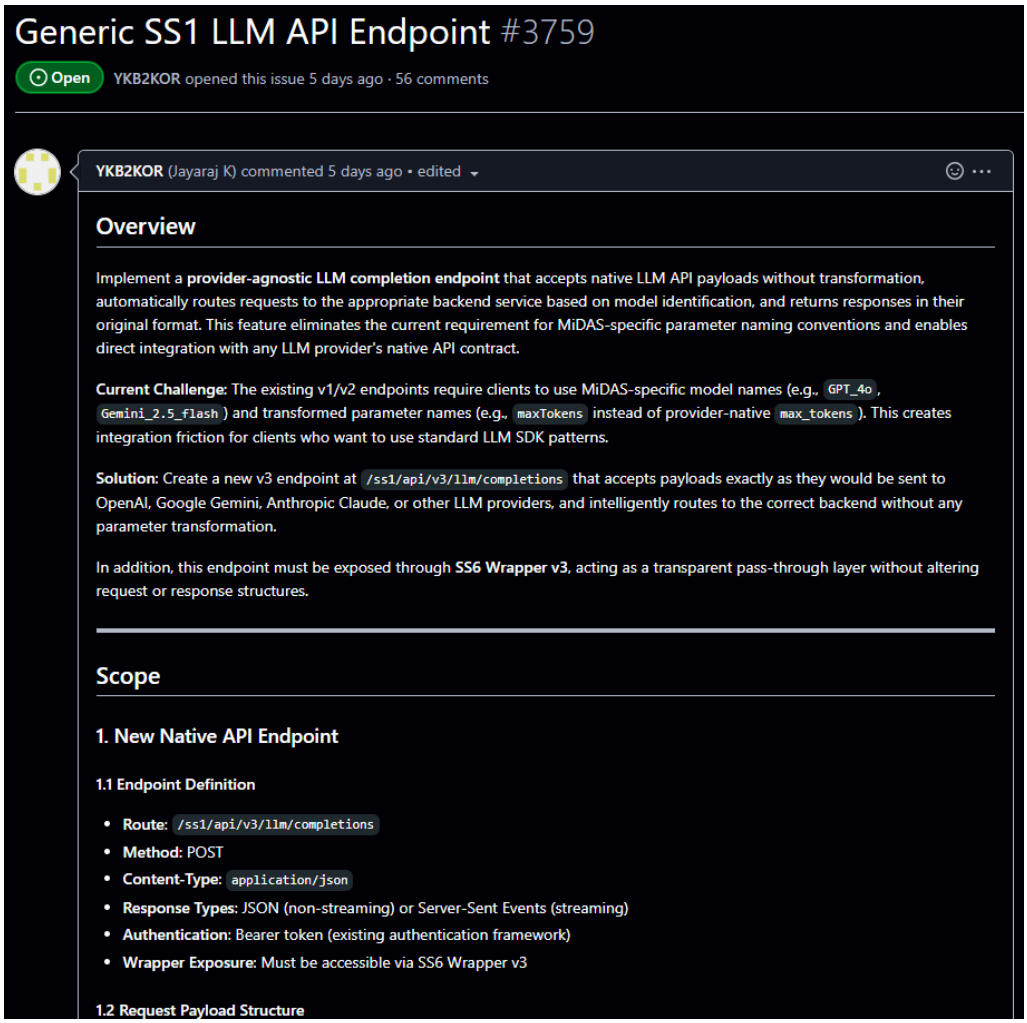
How to Use the SDD Agent

The SDD Agent is triggered from GitHub comments.

Step 1: Navigate to the GitHub Issue

The issue should include:

- Objective
- Scope
- Constraints
- Acceptance criteria
- Subsystem references (if applicable)



Step 2: Trigger the Agent

In the issue comment section, type:

```
@MiDAS /SDD <your request>
```

Examples:

```
@MiDAS /SDD Build a complete evaluation feature for SS6 with
authentication and logging.
```

```
@MiDAS /SDD Design and plan only for this feature.
```

```
@MiDAS /SDD Implement this feature from the existing design.
```

Step 3: Workflow Orchestration

Based on your request, the SDD Agent determines which phases to execute.

It dynamically fetches the required sub-agent prompts when needed

Sub-prompts are fetched only when required for that phase.

Development Lifecycle Phases

The SDD Agent operates across eight structured phases.

Phase 1 – Requirements Analysis

- Analyze user request
- Define functional and non-functional requirements
- Clarify ambiguities
- Document acceptance criteria

Output: Structured requirement specification

Phase 2 – Design

- Fetch design sub-prompt
- Create architecture documentation
- Define APIs and components
- Document technical decisions

Output: Comprehensive design document

Phase 3 – Planning

- Fetch planning sub-prompt
- Break work into phases
- Identify dependencies
- Assess complexity and risks

Output: Step-by-step implementation plan

Phase 4 – Implementation

- Fetch implementation sub-prompt
- Write code for MiDAS development activity (Usecase and Platform)
- Follow MiDAS coding standards
- Ensure proper validation and error handling

Output: Working feature implementation

Phase 5 – Testing

- Fetch testing sub-prompt
- Create unit and integration tests

- Validate edge cases
- Ensure adequate coverage

Output: Passing test suite

Phase 6 – Code Review

- Fetch code review sub-prompt
- Identify quality gaps
- Validate security and performance
- Provide severity-based recommendations

Output: Structured code review report

Phase 7 – Documentation

- Update README files
- Document API endpoints
- Add inline documentation
- Create user-facing documentation if required

Output: Complete documentation updates

Phase 8 – Deployment Readiness

- Verify test success
- Validate configurations
- Confirm environment setup
- Check database migrations
- Prepare deployment checklist

Output: Deployment-ready feature

Workflow Modes

Full SDD Workflow

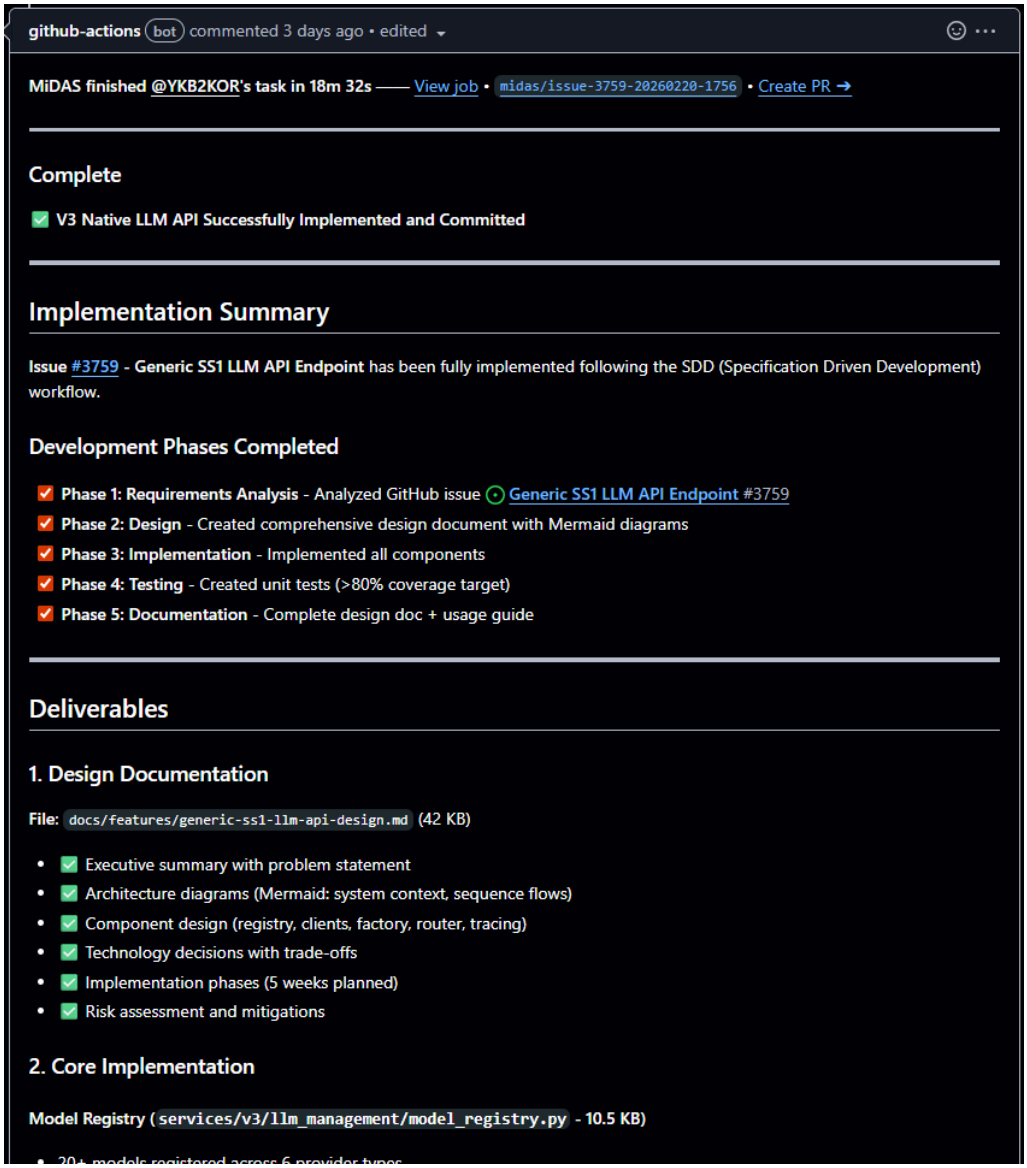
- Requirements → Design → Planning → Implementation → Testing → Code Review → Documentation → Deployment Readiness

Partial Workflow

- Executes only requested or pending phases

Adaptive Execution

- Stops and clarifies when requirements are ambiguous
- Adjusts dynamically based on project context



github-actions bot commented 3 days ago • edited

MiDAS finished @YKB2KOR's task in 18m 32s —— View job • midas/issue-3759-20260220-1756 • Create PR ➔

Complete

✅ V3 Native LLM API Successfully Implemented and Committed

Implementation Summary

Issue #3759 - **Generic SS1 LLM API Endpoint** has been fully implemented following the SDD (Specification Driven Development) workflow.

Development Phases Completed

- ✅ **Phase 1: Requirements Analysis** - Analyzed GitHub issue Generic SS1 LLM API Endpoint #3759
- ✅ **Phase 2: Design** - Created comprehensive design document with Mermaid diagrams
- ✅ **Phase 3: Implementation** - Implemented all components
- ✅ **Phase 4: Testing** - Created unit tests (>80% coverage target)
- ✅ **Phase 5: Documentation** - Complete design doc + usage guide

Deliverables

1. Design Documentation

File: `docs/features/generic-ss1-llm-api-design.md` (42 KB)

- ✅ Executive summary with problem statement
- ✅ Architecture diagrams (Mermaid: system context, sequence flows)
- ✅ Component design (registry, clients, factory, router, tracing)
- ✅ Technology decisions with trade-offs
- ✅ Implementation phases (5 weeks planned)
- ✅ Risk assessment and mitigations

2. Core Implementation

Model Registry (`services/v3/llm_management/model_registry.py` - 10.5 KB)

- 20+ models registered across 6 provider types

AC13	Documentation	✅	Design + usage guide
AC14	Testing	✅	Unit tests (>80% coverage)
AC15	Backward compatibility	✅	v1/v2 unchanged

Total: 15/15 Acceptance Criteria Met ✅

Code Commit

Commit SHA: `4e6df3d83f0661f29a3373530d40b269aa65a756`

Files Committed (16 files):

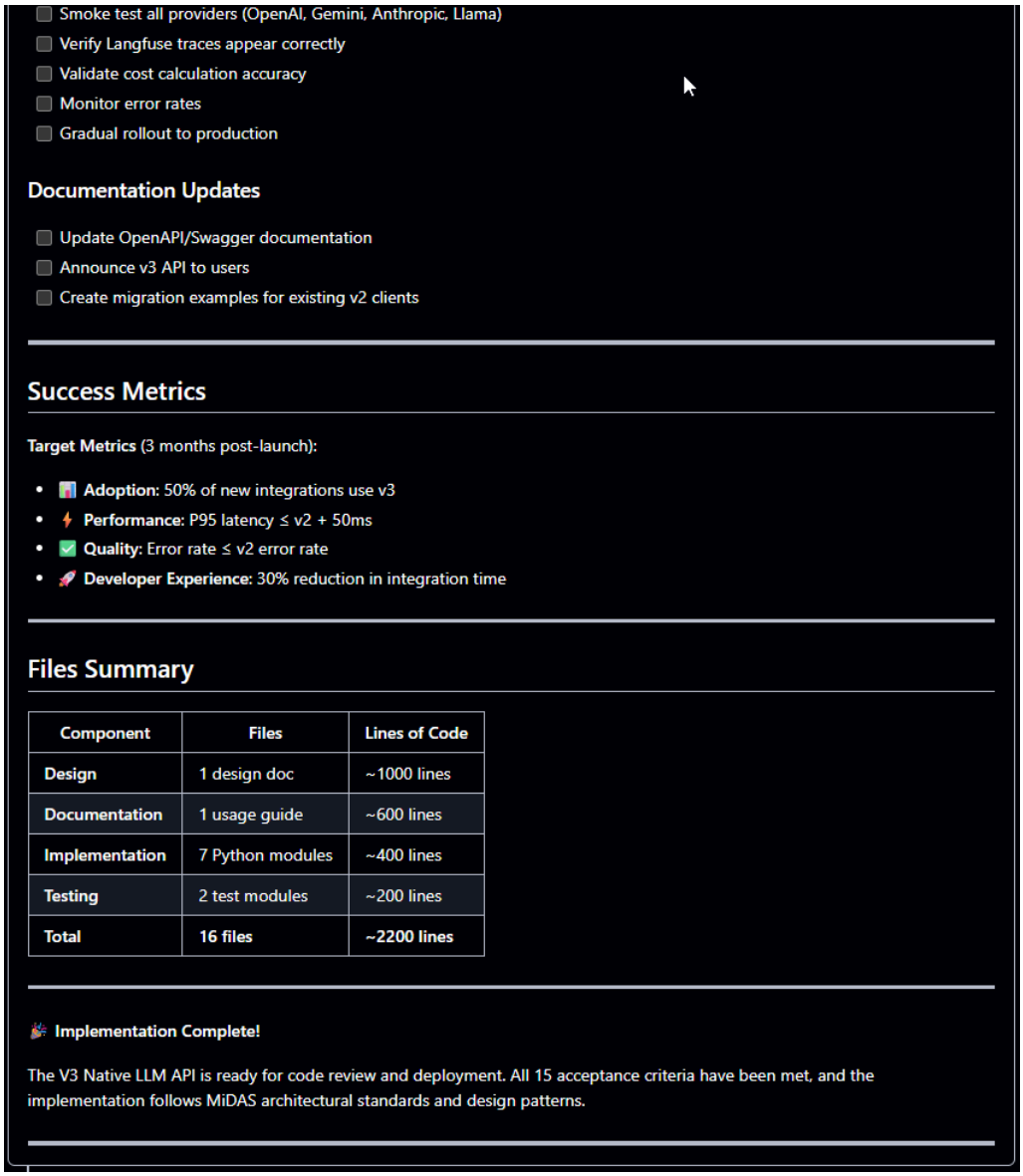
1. `docs/features/generic-ss1-llm-api-design.md` (Design document)
2. `docs/features/v3-native-llm-api-usage-guide.md` (Usage guide)
3. `ss1-llms/app/appmain.py` (Router registration)
4. `ss1-llms/app/api/v3/native_llm/__init__.py`
5. `ss1-llms/app/api/v3/native_llm/datamodels.py`
6. `ss1-llms/app/api/v3/native_llm/llm_router.py`
7. `ss1-llms/app/services/v3/__init__.py`
8. `ss1-llms/app/services/v3/llm_management/__init__.py`
9. `ss1-llms/app/services/v3/llm_management/model_registry.py`
10. `ss1-llms/app/services/v3/llm_management/native_clients.py`
11. `ss1-llms/app/services/v3/llm_management/client_factory.py`
12. `ss1-llms/app/services/v3/tracing/__init__.py`
13. `ss1-llms/app/services/v3/tracing/native_tracing.py`
14. `ss1-llms/tests/unit/services/v3/__init__.py`
15. `ss1-llms/tests/unit/services/v3/test_model_registry.py`
16. `ss1-llms/tests/unit/services/v3/test_client_factory.py`

Commit Message:

```
feat: Implement V3 Native LLM API with provider-agnostic routing

Implements #3759 - Generic SS1 LLM API Endpoint
[Full implementation details in commit message]

Co-Authored-By: Claude Sonnet 4.5 <noreply@anthropic.com>
```



Quality Standards

The SDD Agent enforces the following:

- Code follows SOLID principles
- Proper type hints and docstrings
- Structured error handling
- 80%+ test coverage
- No critical code review issues

- Clear documentation

Success Criteria

A feature is considered complete when:

- Requirements are clearly defined
- Design is validated
- Implementation follows standards
- Tests pass successfully
- Code review identifies no critical issues
- Documentation is complete
- Feature is ready for deployment

Best Practices

- Provide complete and structured requirements.
- Specify which phases should be executed if partial workflow is desired.
- Use the SDD Agent for complex or cross-subsystem features.
- Review outputs at each phase before proceeding.
- Treat it as a lifecycle orchestrator, not a single-purpose agent.

Limitations

- The agent relies on clarity of requirements.
- It does not deploy directly to production.
- It assumes repository conventions unless specified.
- Human review and approval are still required before merge or release.

Summary

The Specification Driven Development (SDD) Agent provides structured, lifecycle-driven orchestration for feature development across the MiDAS platform.

It ensures:

- Specification-first thinking
- Structured planning
- High-quality implementation
- Comprehensive validation
- Deployment readiness

It should be used for complete feature execution where quality, structure, and lifecycle discipline are required.

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